Wenxuan Guo

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Research interests

Experimental Design: optimal experimental design, covariate balance
Causal Inference: randomization inference, design-based inference
Optimization and Decision-Making: online learning, optimal transport

Education

University of Chicago, Booth School of Business Chicago, IL
Ph.D. in Econometrics and Statistics, expected 2027 Sep. 2021 – Present
Advisors: Tengyuan Liang and Panos Toulis

University of Chicago, Department of Statistics Chicago, IL
M.S. in Statistics Sep. 2019 – Jul. 2021
Advisor: Lek-Heng Lim

Fudan University, School of Mathematical Sciences Shanghai, China
B.S. in Mathematical Sciences Sep. 2015 – Jul. 2019
Advisor: Shuqin Zhang

Job market paper

Experimental Design When N Equals One [Paper]
Wenxuan Guo and Tengyuan Liang. 2026.
Major Revision, Biometrika.

Publications and manuscripts

ML-assisted Randomization Tests for Detecting Treatment Effects in A/B Experiments [Paper]
Wenxuan Guo, JungHo Lee, and Panos Toulis. 2026.
Advances in Neural Information Processing Systems (NeurIPS).

Gaussianized Design Optimization for Covariate Balance in Randomized Experiments [Paper]
Wenxuan Guo, Tengyuan Liang, and Panos Toulis. 2026.
Journal of the Royal Statistical Society Series B (JRSS-B).
An earlier version appeared as an extended abstract in the *Proceedings of the ACM Conference on Economics and Computation (EC 2025).*

Reversible Gromov-Monge Sampler for Simulation-Based Inference [Paper]
YoonHaeng Hur, **Wenxuan Guo**, and Tengyuan Liang. 2024.
SIAM Journal on Mathematics of Data Science (SIMODS).

Online Learning to Transport via the Minimal Selection Principle [Paper]
Wenxuan Guo, YoonHaeng Hur, Tengyuan Liang, and Christopher Thomas Ryan. 2022.
Conference on Learning Theory (COLT).

Permutation Inference under Multi-way Clustering and Missing Data [Paper]
Wenxuan Guo, Panos Toulis, and Yuhao Wang. 2026.
Submitted.

Invariance-based Inference in High-Dimensional Regression with Finite-Sample Guarantees [Paper]
Wenxuan Guo and Panos Toulis. 2024+.
Reject & Resubmit, Journal of the American Statistical Association (JASA).

Selected research in progress

A Coupling Approach to Cost-Aware Online Portfolio Rebalancing
Wenxuan Guo and Tengyuan Liang.

Statistics of the Shepp p-Product
Wenxuan Guo and Lek-Heng Lim.

Awards and honors

Silver Reviewer Award, International Conference on Machine Learning (ICML)	2026
Master's Program Scholarship, Department of Statistics, University of Chicago	2019 – 2021
Meritorious Winner, Mathematical Contest in Modeling	2019
Selected Member, Top-Notch Student Training Program, School of Mathematical Sciences, Fudan University	2018
Zengshou Xu Scholarship, Fudan University (top 10%)	2018
First Place, Contemporary Undergraduate Mathematical Contest in Modeling	2017
Freshman Scholarship, Fudan University (top 5%)	2015

Software

R package "`mwperm`" (under development)
Provides finite-sample valid permutation inference for linear models under multi-way clustering, including dyadic and panel-data settings.
GitHub "`gdo_paper`"
Provides an R implementation of Gaussianized Design Optimization (GDO), a novel method for achieving optimal covariate balance in randomized experiments.

Teaching experience

Booth School of Business, University of Chicago

BUSN 41903: Applied Econometrics (Ph.D. core)
– Assistant Instructor 2024 – 2026

BUSN 41000: Business Statistics (MBA core)
– Assistant Instructor 2024 – 2026

BUSN 41910: Time-series Analysis for Forecasting and Model Building (Ph.D. core)
– Teaching Assistant 2025

Department of Statistics, University of Chicago

STAT 34800: Modern Applied Optimization (graduate-level)
– Teaching Assistant 2023

STAT 32940: Multivariate Data Analysis via Matrix Decompositions (graduate-level)
– Teaching Assistant 2020

Talks

Experimental Design When N Equals One

Econometrics & Statistics Ph.D. Seminar, Chicago, IL Oct. 2026
Carnegie Mellon University (CMU) Causal Group (Online) Sep. 2026

Permutation Inference under Multi-way Clustering and Missing Data

Econometrics & Statistics Ph.D. Seminar, Chicago, IL Mar. 2026
Chicago Chinese Social Research Group (CCSRG), Chicago, IL Feb. 2026

Gaussianized Design Optimization for Covariate Balance in Randomized Experiments

2026 INFORMS Annual Meeting, San Francisco, CA Nov. 2026
Session Title: Experimentation
Joint Conference on Statistics and Data Science in China (JCSDS 2026), Guizhou, China Jul. 2026
American Causal Inference Conference (ACIC 2026), Salt Lake City, UT May 2026
Advances with Field Experiments Conference (AFE 2025), Chicago, IL Sep. 2025
Joint Statistical Meetings (JSM 2025), Nashville, TN Aug. 2025
ACM Conference on Economics and Computation (EC 2025), Palo Alto, CA Jul. 2025
Econometrics & Statistics Ph.D. Seminar, Chicago, IL Apr. 2025

ML-assisted Randomization Tests for Complex Treatment Effects in A/B Experiments

American Causal Inference Conference (ACIC 2025), Detroit, MI May 2025
Online Causal Inference Seminar (OCIS) Dec. 2024
Discussant: Professor Xinran Li (University of Chicago)

Invariance-based Inference in High-Dimensional Regression with Finite-Sample Guarantees

Joint Conference on Statistics and Data Science in China (JCSDS), Yunnan, China — Jul. 2024

Econometrics & Statistics Ph.D. Seminar, Chicago, IL — Jan. 2024

Academic service

Reviewer for NeurIPS, ICML, AAAI, *Journal of the Royal Statistical Society Series B* (JRSS-B), *Statistica Sinica*, and *Scandinavian Journal of Statistics*.

Initiated and organized the ChicagoBooth causal inference reading group (2025–2026)

Industry experience

Google — San Bruno, CA

Data Science Research Intern on Experimentation — May – Aug. 2025

- Developed and integrated efficient rerandomization experiments to improve covariate balance.
- Implemented bias-minimizing experimental designs for more reliable large-scale hyperparameter tuning.
- This experience motivated the subsequent paper, "*Experimental Design When N Equals One*," which studies optimal design for time-series experiments.